

D1.0 DEC Tour Guide: Initial CSSR4Africa ROS1 Prototype

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Responsible Person: **Yohannes Haile**

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Executive Summary

Deliverable D1.0 documents the completion of Tasks 1.1 and 1.2 for the Digital Experience Center (DEC) Pepper robot project at Carnegie Mellon University Africa. Task 1.1 covers the creation of an operational navigation map of the DEC space and a structured knowledge base encoding the location, pose, and exhibit information for each tour stop. Task 1.2 covers the design of a tour scenario script and its realisation as an XML behavior tree specification that is interpreted at runtime by the D5.4.3 Robot Mission Interpreter. Together, these two artifacts form the data foundation on which the autonomous DEC tour runs: the map enables safe, goal-directed navigation, while the knowledge base and behavior tree define what the robot does at each point along the tour.

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1 Introduction

The Digital Experience Center (DEC) at Carnegie Mellon University Africa hosts a set of exhibit booths that showcase research in Digital Public Infrastructure (DPI), Digital Public Goods (DPGs), cybersecurity, and related areas. A central goal of the DEC project is to equip a Pepper humanoid robot with the ability to conduct autonomous, visitor-facing tours of this space — detecting visitors, navigating to each exhibit, presenting relevant information, handling questions, and concluding the tour without human intervention.

Two foundational prerequisites must be in place before any tour can run:

1. **A spatial map and a location knowledge base.** The robot requires a metric map of the physical DEC environment in order to plan collision-free paths between booths. In addition, it needs a structured knowledge base that associates each tour stop with the exact robot pose for that location, the coordinates of gesture targets at the exhibit, and the spoken content to deliver there.
2. **A scenario script and a behavior tree.** The sequence of actions that constitute a tour — visitor detection, greeting, consent gathering, exhibit visits, question handling, and farewell — must be specified in a form that the robot's Behavior Controller can interpret and execute. This is accomplished through a natural-language scenario script that is then formalised into an XML behavior tree specification.

Both tasks build directly on the **CSSR4Africa** project, which provides the ROS1-based software stack for the Pepper robot including a CAD-based map generation module, a navigation stack, and a script interpreter for behavior tree execution [1]. The DEC work reuses these ROS1 components for environment mapping and adapts the CSSR4Africa behavior tree format for the DEC tour scenario.

This deliverable documents the work carried out under Tasks 1.1 and 1.2 to produce these two artifacts.

2 Requirements Definition

Task 1.1 — Map and Knowledge Base

- A 2D occupancy-grid map of the DEC must be generated that accurately represents the floor plan, booth locations, and all fixed obstacles in the space.
- A knowledge base must record, for each tour stop: a unique location identifier, a descriptive name, the robot's target pose (position and orientation), gesture target coordinates (used by the Behavior Controller for deictic pointing), and the exhibit speech content.
- The knowledge base must be stored in a human-readable, version-controllable format (.dat) and loaded at runtime by the Behavior Controller without requiring a rebuild of any package.
- A culture knowledge base must supply language-specific utility phrases (greetings, farewells, consent requests) so that tour content can be adapted for different visitor audiences without modifying the behavior tree.

Task 1.2 — Scenario Script and Behavior Tree

- A plain-language scenario script must enumerate every distinct step of the DEC tour in the order in which it occurs, specifying the actor (robot or visitor), the action, and any decision branches.
- The scenario script must cover visitor detection, engagement, tour consent, exhibit navigation and presentation, visitor question handling, and tour conclusion.
- The scenario must be translated into an XML behavior tree specification that can be interpreted by the script interpreter developed in CSSR4Africa, and subsequently executed at runtime by the D5.4.3 Robot Mission Interpreter without manual code changes.
- The behavior tree must use the custom BT node vocabulary defined in the D5.4.3 Robot Mission Interpreter: action nodes for navigation, speech, gesture, and conversation; service nodes for attention and speech control; and stateful nodes for face detection and speech listening.

3 Task 1.1 — Map and Knowledge Base for DEC

3.1 DEC Navigation Map

Mapping Process

The DEC environment map was generated using the **CSSR4Africa** `mapGeneration` ROS1 node, documented in CSSR4Africa Deliverable D5.5.3 [2]. Rather than a SLAM-based approach, the module uses a **CAD-based method**: geometric descriptions of the DEC floor plan are provided as `.dat` input files, and the node processes them to produce a metric occupancy grid without requiring the robot to physically explore the environment.

Two `.dat` input files drive the process:

- `mapGenerationInput.dat` — specifies the map dimensions (width and height in metres), the map resolution (metres per pixel), and the filenames for the obstacle definition file and the two output maps.
- `obstacles.dat` — lists each obstacle as a geometric primitive in the form `OBSTACLE X Y WIDTH HEIGHT`, defining the rectangular booths, walls, and fixed furniture of the DEC space.

The node parses these primitives and rasterises them onto an occupancy grid using Bresenham's line algorithm for edges, flood-fill for solid interiors, and polygon-filling for complex shapes.

Output Maps

The `mapGeneration` node produces two PNG output files, each also exported as a ROS `OccupancyGrid`:

- `environmentMap.png` — the workspace map representing the physical layout of the DEC (free space and obstacles).
- `configurationSpaceMap.png` — the configuration space map, derived by applying morphological image dilation (via OpenCV `cv::dilate`) with a circular structuring element whose radius matches the Pepper robot's base. This inflation ensures the planner keeps the robot safely clear of all obstacles throughout the space.

Figure 1 shows the workspace map generated for the DEC space.

3.2 Environment Knowledge Base

The environment knowledge base is a structured `.dat` file that serves as the single source of truth for all location-specific data consumed during the tour. Each entry is keyed by a unique location identifier and contains the fields summarised in Table 1.

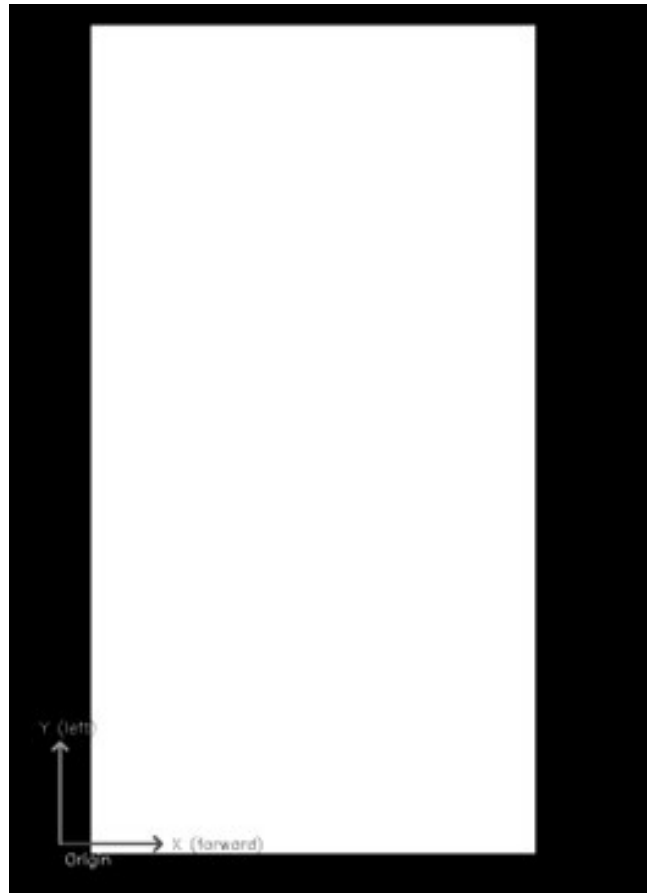


Figure 1: Occupancy grid workspace map of the DEC space generated by the CSSR4Africa mapGeneration node from CAD data. White regions represent navigable free space; black regions represent walls, booths, and fixed obstacles. The coordinate axes at the bottom-left indicate the map origin and orientation.

Field	Description
name	Human-readable name of the tour stop or exhibit booth.
description	Brief text description of the exhibit shown at this location.
robot_pose.x, y, theta	Target position and orientation (in metres and radians) that the robot navigates to when visiting this stop.
gesture_target.x, y, z	3D coordinates (in metres, map frame) of the exhibit panel or object toward which the robot performs a deictic pointing gesture.
exhibit_speech	The spoken text delivered by the robot upon arriving at the exhibit. Consumed by the TTS action node.

Table 1: Fields defined for each location entry in the environment knowledge base.

Tour Specification

In addition to individual location entries, the knowledge base includes a tour specification — an ordered list of location identifiers that defines the sequence in which the robot visits the exhibit booths.

The Behavior Controller's `RetrieveListOfExhibits` BT node reads this list at the start of the tour and places it onto the blackboard as the `exhibit_queue`. The tour loop then pops one entry at a time, navigating to and presenting each exhibit in order.

Two variants of the environment knowledge base are maintained: a full version containing all DEC exhibit stops, and a shorter version containing a subset of stops used during development and testing.

3.3 Culture Knowledge Base

The culture knowledge base is a separate `.dat` file that supplies language-tagged utility phrases — greetings, farewell messages, tour invitation text, and consent-request sentences. Phrases are indexed by a phrase identifier and a language tag (e.g. `English`). The Behavior Controller's `KnowledgeManager` singleton retrieves the appropriate phrase at runtime via `getUtilityPhrase(phraseId, language)`, with the active language controlled by the `language` key in the configuration file. This design allows the tour to be adapted for additional languages without modifying any C++ code or behavior tree XML.

4 Task 1.2 — Scenario Script and Behavior Tree for DEC Tour

4.1 Tour Scenario Script

The scenario script describes the DEC tour as an ordered sequence of discrete events. Each event specifies the responsible actor (robot or visitor), the action performed, and any conditional branches that determine the subsequent flow. The script is structured around five high-level phases:

1. **Visitor Detection.** The robot waits in an idle state, running a looping background animation, while continuously monitoring the face-detection stream. Once a visitor is detected within the configured timeout, the detection phase succeeds.
2. **Visitor Engagement.** The robot stops its idle animation, executes a welcoming gesture, and delivers a greeting utterance. It then attempts to establish mutual gaze. If mutual gaze is not achieved within the timeout, the robot falls back gracefully and returns to the detection phase.
3. **Exhibit Visits.** The robot retrieves the ordered exhibit list from the knowledge base and enters a loop. For each exhibit, it navigates to the target pose, performs a deictic gesture toward the exhibit panel, delivers the exhibit speech before advancing to the next stop.
4. **Tour Conclusion.** After all exhibits have been visited, the robot delivers a farewell utterance, disables overt attention, and returns to the idle detection state.

The script also specifies two cross-cutting concerns: the Speech Recognition node is muted during all TTS playback to prevent the robot from transcribing its own speech; and the Overt Attention node is kept active throughout the tour to maintain natural head-tracking behavior toward the visitor.

4.2 Behavior Tree Specification

The scenario script is formalised as an XML behavior tree specification. The behavior tree format follows the schema used by the **CSSR4Africa** script interpreter, which was developed as part of the CSSR4Africa ROS1 software stack to interpret structured XML scenario files and drive the Pepper robot through a sequence of actions. For the DEC tour, the scenario XML was authored to be compatible with the CSSR4Africa interpreter, enabling the existing ROS1 infrastructure to be reused directly for initial validation. The tree is also loaded and executed at runtime by the D5.4.3 Robot Mission Interpreter, which implements an equivalent interpreter using BehaviorTree.CPP v4 [3]. The tree is structured into five subtrees, one per tour phase, as described in Table 2.

Subtree	Description
I. DetectVisitor	A Parallel node simultaneously enables the animate behavior (SetAnimateBehavior: enabled) and sets overt attention to scanning mode (SetOvertAttentionMode: scanning). The sequence then blocks on IsVisitorDiscovered until a face is detected.
II. EngageVisitor	Disables the animation, performs an iconic welcome gesture (PerformIconicGesture: welcome), and re-enables animation. Delivers a welcome speech (SayText 4), sets attention to seeking mode, blocks on IsMutualGazeDiscovered, switches to social attention mode, then delivers the tour invitation speech (SayText 5).
III. QueryVisitorResponse	First checks IsASREnabled. If ASR is unavailable, retries up to three times: prompts the visitor with a yes/no speech (SayText 1) and waits for a button press (PressYesNoDialogue). If ASR is available, enables the speech event, captures the visitor utterance via GetVisitorResponse, and disables the speech event. In both paths the response is evaluated by IsVisitorResponseYes; a refusal triggers a polite decline speech (SayText 3) and forces FAILURE to abort the tour.
IV. VisitExhibit	Calls RetrieveListOfExhibits and ResetRobotPose, then enters a KeepRunningUntilFailure loop (inverted to return SUCCESS on completion): while IsListWithExhibit succeeds, SelectExhibit loads the next stop, a follow-me speech (SayText 2) is delivered, the helper subtree _NavigateToLocation drives the robot to the exhibit, overt attention is set to social mode, the pre-gesture exhibit speech is delivered (DescribeExhibitSpeech 1), attention switches to location mode for PerformDeicticGesture, then attention returns to social mode for the post-gesture speech (DescribeExhibitSpeech 2). The loop exits when the exhibit list is exhausted.
_NavigateToLocation	Helper subtree called by IV. VisitExhibit. Disables overt attention and animate behavior, executes Navigate to the target pose, re-enables attention in seeking mode, and blocks on IsMutualGazeDiscovered to reestablish eye contact with the visitor after arrival.
V. EndTour	Delivers a farewell speech (SayText 7) and performs an iconic goodbye gesture (PerformIconicGesture: goodbye).

Table 2: Subtrees of the main TourGuide behavior tree.

4.3 Composite Node Structure

Every action in the tree is wrapped in a Fallback paired with a HandleFallBack node, so any single-node failure is caught and handled gracefully without aborting the enclosing Sequence. The five phase subtrees are chained under a top-level Sequence rooted at TourGuide, preceded by a StartOfTree initialisation node.

The exhibit loop in IV. VisitExhibit uses a KeepRunningUntilFailure composite wrapped in an Inverter: when IsListWithExhibit finally returns FAILURE (queue empty), KeepRunningUntilFailure propagates that failure, and the Inverter converts it to SUCCESS, allowing the outer Sequence to proceed to V. EndTour.

The III. QueryVisitorResponse subtree implements a dual input path: if ASR is disabled (e.g. in a noisy environment), a RetryUntilSuccessful loop presents a physical button-press dialogue up to three times before the tree continues. If ASR is available, the sequence bypasses the button path and captures speech directly. On visitor refusal, a ForceFailure node ensures the overall TourGuide sequence terminates cleanly after delivering the decline message.

4.4 BT Node Catalogue

Table 3 lists all custom nodes registered in the TreeNodesModel of the behavior tree XML.

Node ID	Type	Description
StartOfTree	Action	Initialisation node executed once at the start of the TourGuide tree.
HandleFallback	Action	Fallback handler invoked whenever a sibling action fails; logs the failure and returns SUCCESS to prevent cascading aborts.
IsASREnabled	Condition	Returns SUCCESS if the ASR subsystem is active and available for speech input.
IsVisitorDiscovered	Condition	Returns SUCCESS when the face-detection subsystem reports at least one person present.
IsMutualGazeDiscovered	Condition	Blocks until sustained mutual gaze is established between the visitor and the robot.
IsVisitorResponseYes	Condition	Returns SUCCESS if the visitor's response (ASR or button) is affirmative; FAILURE otherwise.
IsListWithExhibit	Condition	Returns SUCCESS while the exhibit queue is non-empty; FAILURE when all exhibits have been visited.
SetAnimateBehavior	Action	Enables or disables the looping background animation. Port: name (enabled / disabled).
SetOvertAttentionMode	Action	Configures the overt attention module. Port: name (scanning / seeking / social / location / disabled).
SetSpeechEvent	Action	Enables or disables the speech recognition event listener. Port: name (enabled / disabled).
PerformIconicGesture	Action	Executes a full-body iconic gesture. Port: name (welcome / goodbye).
PerformDeicticGesture	Action	Points toward the gesture target coordinates loaded for the current exhibit by SelectExhibit.
SayText	Action	Delivers a pre-defined speech phrase by index. Port: name (phrase ID, e.g. 1-7).
DescribeExhibitSpeech	Action	Delivers exhibit-specific speech loaded from the knowledge base. Port: name (1 = pre-gesture speech; 2 = post-gesture speech).
GetVisitorResponse	Action	Listens for a visitor utterance via ASR and writes the result to the blackboard.
PressYesNoDialogue	Action	Displays a physical yes/no button dialogue on the robot's tablet and waits for a press.
RetrieveListOfExhibits	Action	Reads the tour specification from the knowledge base and populates the exhibit queue on the blackboard.
ResetRobotPose	Action	Returns the robot to the home pose before beginning the exhibit loop.
SelectExhibit	Action	Reads the next entry from the exhibit queue and loads its speech, navigation goal, and gesture target onto the blackboard.
Navigate	Action	Sends the current exhibit's navigation goal to the motion planning stack and blocks until arrival.

Table 3: Custom BT nodes defined in the TourGuide behavior tree XML.

Figure 2 shows the Pepper robot executing the tour behavior tree during a live demonstration in the DEC space, performing a deictic gesture toward the DPG/DPI Booth 2 exhibit.



Figure 2: Pepper robot performing a deictic gesture at DPG/DPI Booth 2 during a live DEC tour demonstration. The robot has navigated to the booth pose defined in the environment knowledge base and is directing the visitor's attention to the exhibit.

5 Knowledge Base Documentation

The tables in this section document the nine tour stops defined in the environment knowledge base .data file produced under Task 1.1. Table 4 provides the spatial data for all stops. Tables 5–13 list the multilingual speech content delivered at each stop, divided into a *pre-gesture message* (delivered upon arrival, before the deictic gesture) and a *post-gesture message* (delivered after the gesture, typically addressed to the human researcher at the booth).

Tour Stop Spatial Summary

#	Description	Robot Pose (x, y, θ°)	Gesture Target (x, y, z)	Cultural Knowledge
1	Pepper's starting location	(4.0, 6.8, 90°)	(0.0, 0.0, 0°)	SymbolicShapeRespect, BowExtentRespect
2	DPG/DPI Booth 1	(4.8, 8.0, 45°)	(6.0, 8.0, 0°)	DeicticShape
3	DPG/DPI Booth 2	(4.8, 5.6, 45°)	(6.0, 5.6, 0°)	DeicticShape
4	Data Exchange Booth 1	(3.2, 1.6, 100°)	(3.2, 0.4, 270°)	DeicticShape
5	Data Exchange Booth 2	(1.6, 1.6, 50°)	(1.6, 0.4, 270°)	DeicticShape
6	Connectivity Booth	(2.4, 3.2, 50°)	(1.2, 3.2, 180°)	DeicticShape
7	Cybersecurity Booth 1	(2.4, 5.2, 50°)	(1.2, 5.2, 180°)	DeicticShape
8	Cybersecurity Booth 2	(2.4, 7.6, 50°)	(1.2, 7.6, 180°)	DeicticShape
9	Pepper's ending location	(4.0, 6.8, 100°)	(0.0, 0.0, 0°)	IconicShape

Table 4: Spatial summary of the nine DEC tour stops. Pose values are in metres (x, y) and degrees (θ). Gesture target z gives the horizontal bearing in degrees toward the exhibit panel; z = 0 indicates no deictic gesture at that stop.

Tour Stop Speech Content

Language	Pre-Gesture Message	Post-Gesture Message
English	Hello and thank you for visiting the Upanzi DEC. I'm your guide. In this tour, you will be the citizen protagonist.	Together, we'll explore how Upanzi's open-source digital solutions make everyday services easier and more secure. Feel free to ask questions as we go along.
isiZulu	<i>No message defined.</i>	<i>No message defined.</i>
Kinyarwanda	Murakazaneza muri laboratwari ya robotikisi muri kaneji meloni inivarisite yafurika.	Ngiye kubatemberezaho gato, nizereko muribuze kuryoherwa.

Table 5: Speech content for Stop 1: Pepper's starting location.

Language	Pre-Gesture Message	Post-Gesture Message
English	Our first stop is crucial – establishing your digital identity. Here a researcher will get you registered for a National Digital ID.	Please introduce yourself, and help our citizen out.
isiZulu	<i>No message defined.</i>	<i>No message defined.</i>
Kinyarwanda	Iyi yitwa pepa humanoyidi roboti.	Tuyikoresha mubushakashatsi mubijyanye nimibanire hamwe nimikoranire hagati yabantu ndetse naza robo.

Table 6: Speech content for Stop 2: DPG/DPI Booth 1.

Language	Pre-Gesture Message	Post-Gesture Message
English	Welcome to the Bank of Upanzi! Our able bank officer will assist you in opening your digital bank account today.	Please take over and help our citizen out.
isiZulu	<i>No message defined.</i>	<i>No message defined.</i>
Kinyarwanda	Iyi yitwa pepa humanoyidi roboti.	Tuyikoresha mubushakashatsi mubijyanye nimibanire hamwe nimikoranire hagati yabantu ndetse naza robo.

Table 7: Speech content for Stop 3: DPG/DPI Booth 2.

Language	Pre-Gesture Message	Post-Gesture Message
English	At this station a Program Officer helps citizens enrol in social benefit programs. In our scenario, Upanzi Republic offers a fertiliser subsidy to farmers, and you will sign up for it.	This process uses the Upanzi Programs Management System (UPMS). Please help out our citizen.
isiZulu	<i>No message defined.</i>	<i>No message defined.</i>
Kinyarwanda	Iyi robo ni likisi moshoni.	Tuyikoresha mukwigisha roboti manipileshoni.

Table 8: Speech content for Stop 4: Data Exchange Booth 1.

Language	Pre-Gesture Message	Post-Gesture Message
English	Finally, you arrive at a mock retail shop – MojaShop, a demo point-of-sale that showcases how digital payments and subsidies work in real time.	Our Vendor will help you buy a bag of fertiliser using your subsidy. Over to you.
isiZulu	<i>No message defined.</i>	<i>No message defined.</i>
Kinyarwanda	Nizereko mwaryohewe no gutemberezwa.	Tuzongere kubonana ubutaha.

Table 9: Speech content for Stop 5: Data Exchange Booth 2.

Language	Pre-Gesture Message	Post-Gesture Message
English	Access to the internet is the gateway to digital services. In the Connectivity section, Upanzi showcases projects aimed at expanding and measuring internet access in Africa, especially in underserved rural regions. Currently, a deep urban-rural digital divide persists: in Africa, about 57% of urban dwellers use the Internet, compared to just 23% of people in rural areas. Upanzi's projects tackle this gap from multiple angles – by piloting innovative connectivity solutions, monitoring internet resilience, and promoting open data for informed policy.	Our connectivity engineer will work you through.
isiZulu	<i>No message defined.</i>	<i>No message defined.</i>
Kinyarwanda	Iyiyo yitwa rumba.	Tuyikoresha mukwigisha mobile robotikisi.

Table 10: Speech content for Stop 6: Connectivity Booth.

Language	Pre-Gesture Message	Post-Gesture Message
English	This section showcases Upanzi's cutting-edge cybersecurity research projects. As Africa's digital ecosystem grows – with e-government services, mobile apps, and AI technologies – ensuring their security and integrity is paramount. Upanzi researchers are working on multiple fronts: security testing of e-services and mobile apps, innovative defences against AI-driven threats like deepfake voices, and studying emerging attack vectors such as smishing (SMS phishing).	In this interactive lab, you'll engage with demonstrations of these projects. The goal is to highlight how we proactively find vulnerabilities and develop defences, making digital infrastructure trustworthy for all users.
isiZulu	<i>No message defined.</i>	<i>No message defined.</i>
Kinyarwanda	Iyiyo yitwa rumba.	Tuyikoresha mukwigisha mobile robotikisi.

Table 11: Speech content for Stop 7: Cybersecurity Booth 1.

Language	Pre-Gesture Message	Post-Gesture Message
English	Digital transformation isn't just about technology; it's about people. Upanzi's Capacity Building initiatives focus on empowering individuals – students, researchers, policymakers and implementers – with the skills, knowledge, and ethical frameworks to drive and sustain digital innovation. In this final section of the tour, you will learn about several programs: our annual Digital Identity Hackathon and picoCTF Africa, the use of AI for Policy Analysis, our efforts in Responsible Research and Innovation (RRI), and the aforementioned Data Quality Measurement project. These initiatives ensure that the digital public goods we develop are used effectively and responsibly, and that a pipeline of talent and ideas continues to grow. The format here is part showcase, part dialogue – you'll meet an Innovation Coach	who oversees these programs and even try a mini hackathon challenge yourself!
isiZulu	<i>No message defined.</i>	<i>No message defined.</i>
Kinyarwanda	Iyiyo yitwa rumba.	Tuyikoresha mukwigisha mobile robotikisi.

Table 12: Speech content for Stop 8: Cybersecurity Booth 2 (Capacity Building).

Language	Pre-Gesture Message	Post-Gesture Message
English	Thank you for joining us today and accepting to play the role of citizen protagonist. Overall, the tour painted a comprehensive picture of the Upanzi Network's approach: building interoperable Digital Public Infrastructure that is open-source and scalable, while simultaneously investing in the governance, security, and capacity required to make that infrastructure impactful. You have seen how a citizen's single digital identity threads through various services – a powerful illustration of the whole being greater than the sum of its parts. By the end of this immersive experience, it's evident that digital public goods do not stand alone; they interlock like pieces of a puzzle, supported by robust connectivity and safeguarded by vigilant cybersecurity measures, all driven forward by skilled and conscientious individuals.	Hope you enjoyed the tour. Goodbye.
isiZulu	<i>No message defined.</i>	<i>No message defined.</i>
Kinyarwanda	Murakazaneza muri laboratwari ya robotikisi muri kaneji meloni iniverisite yafurika.	Ngiye kubatemberezaho gato, nizereko muribuze kuryoherwa.

Table 13: Speech content for Stop 9: Pepper's ending location.

References

- [1] CSSR4Africa Deliverable D6.1: Use Case Implementation. Technical report, Carnegie Mellon University Africa, June 2025. Revision 1.2. Available online: https://cssr4africa.github.io/deliverables/CSSR4Africa_Deliverable_D6.1.pdf.
- [2] CSSR4Africa Deliverable D5.5.3: Environment Map Generation. Technical report, Carnegie Mellon University Africa, June 2025. Revision 1.1. Available online: https://cssr4africa.github.io/deliverables/CSSR4Africa_Deliverable_D5.5.3.pdf.
- [3] CSSR4Africa Deliverable D5.4.2: Robot Mission Language. Technical report, Carnegie Mellon University Africa, May 2025. Revision 1.7. Available online: https://cssr4africa.github.io/deliverables/CSSR4Africa_Deliverable_D5.4.2.pdf.

Principal Contributors

The main authors of this deliverable are as follows (in alphabetical order).

Yohannes Haile, Carnegie Mellon University Africa.

Document History

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First draft.

Yohannes Haile.

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